

Does Internal Geometry Break Before Robot Behavior Does?

A preregistered test on SmolVLA/LIBERO: internal activations replace privileged simulator state for failure prediction but do not beat it, and the decodable orientation direction is not causally used

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Repository: github.com/jimfable/mech-int-vla Report commit: see Appendix F

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Executive summary

Question. A robot *policy*—the neural network that turns camera images, the robot’s joint readings and a text instruction into motor commands—sometimes fails at its task. Do the network’s *internal activations* (the intermediate vectors it computes on the way to its output) tell you that a failure is coming, beyond what you could already know from the policy’s *outputs* and from *privileged simulator state* (the true positions of everything, which a simulator knows but a real robot does not)? If yes, reading the internals of such vision-language-action (VLA) models buys something. If no, it does not—at least not in this form.

What we did. We froze one public policy (SmolVLA fine-tuned on LIBERO), one manipulation task (“pick up the book and place it in the back compartment of the caddy”), and three nested failure predictors: M0 (policy outputs only), M1 (M0 + privileged simulator state), M2 (M1 + internal signals built from a probe for the book’s relative orientation). Every threshold, metric, and analysis step was preregistered and hash-locked before the held-out *Locked Test* split (160 episodes, 8 perturbation conditions) was collected. The evaluator ran exactly once. Cost: 21.7 GPU-hours on one RTX 5090.

Results (Figure 1).

1. **The preregistered primary claim fails, and fails cleanly.** M2 beats M1 in paired log loss by +0.47% (90% cluster-bootstrap CI $[-1.2\%, +2.1\%]$; bar: $\geq 3\%$). The interval excludes the bar, but only by 0.9 percentage points: a 3% lift is unlikely, a 1–2% lift is not ruled out.
2. **Internals *substitute* for privileged state.** M2 beats the output-only baseline M0 by 18.8% in log loss (Δ log loss -0.120 , CI $[-0.187, -0.052]$, both ends favourable); AUROC $0.667 \rightarrow 0.835$. A model that never sees the simulator state does as well as one that does.
3. **The decodable orientation direction is a readout, not a lever.** A linear *probe* can read the book’s orientation out of the activations. But when we *edit* the activations along that probe direction (“patching”), the policy’s action moves *less* than when we edit along a random direction of the same size (median off-target ratio 4.0; bar ≤ 0.25), and in the predicted direction in only 48% of 52 valid cases (CI $[36\%, 62\%]$; bar: CI above 50%). The same holds at larger edit sizes and in every perturbation condition.
4. **No earlier alarm.** At a fixed 10% episode-level false-positive rate, M2 and M1 detect 98% of the 60 failures with identical median lead time (paired difference 0 steps, CI $[0, 0]$).

Takeaway. On this policy and task, everything the internal orientation signal knows about upcoming failure is already in the privileged state, and the direction we can decode is not one the policy acts through. Being able to *read* a variable from a network is not the same as the network *using* it. Per the preregistered decision table this is the “neither lift nor specificity” row: publish the negative result, stop the line.

Confidence. High that the measured numbers are what this protocol produces (single frozen run; every input content-addressed). Moderate that the conclusion generalises: one policy, one task, one layer, 158 valid episodes. Section 8 lists what would change our mind, with rough probabilities.

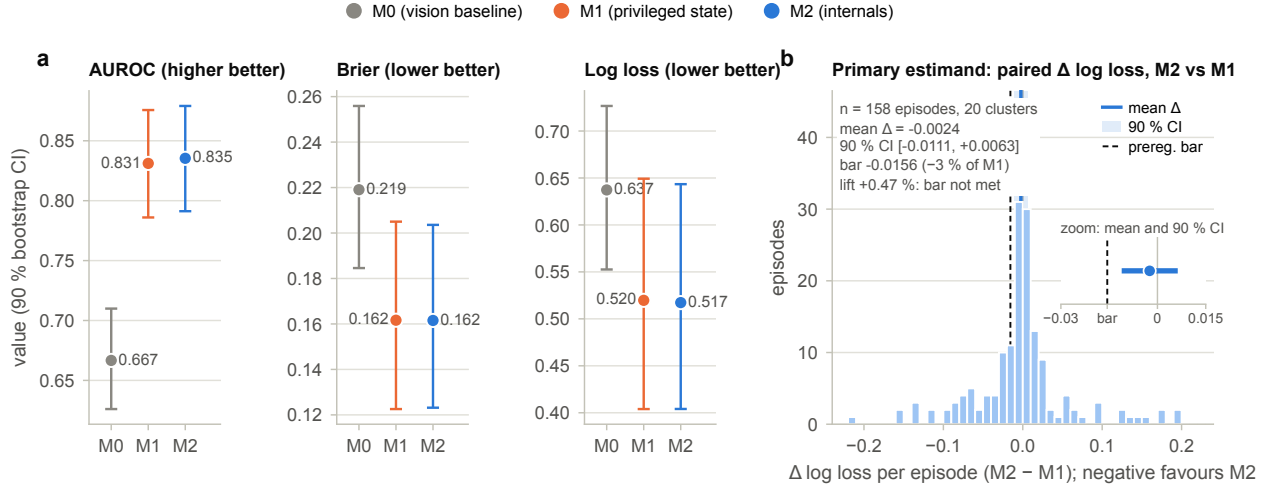


Figure 1: Headline result on the Locked Test (158 valid episodes, 20 init-state clusters). (a) Failure-prediction quality per model with 90% cluster-bootstrap intervals (10 000 replicates, seed 260803). M1 and M2 are indistinguishable; both clearly beat M0. (b) The primary estimand: per-episode paired difference in log loss, $\ell_e^{M2} - \ell_e^{M1}$ (negative = M2 better). Mean -0.0024 , 90% CI $[-0.0111, +0.0063]$; the preregistered bar of a 3% relative improvement corresponds to -0.0156 (dashed) and lies outside the interval. 54% of episodes lie within ± 0.02 . Lower is better for log loss and Brier; higher for AUROC.

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1 The setting in plain terms

This section defines every object the rest of the report uses. Readers who work on VLA models can skip to Section 2.

1.1 The robot and its policy

The robot is a simulated 7-joint arm with a gripper in the LIBERO benchmark. Its **policy** π is a neural network with about 450 million parameters (SmolVLA). At every control step t it receives an *observation* o_t —two camera images (a fixed “agent” view and a wrist camera), the 8 joint/gripper readings, and the text instruction “pick up the book and place it in the back compartment of the caddy”—and outputs a planned sequence of 50 future motor commands, each a 7-dimensional vector (end-effector translation, rotation, gripper). The robot executes only the first command, observes again, and re-plans. So $\pi : o_t \mapsto (a_t, a_{t+1}, \dots, a_{t+49})$, and the closed loop is $o_{t+1} = \text{sim}(o_t, a_t)$.

Internally the network has two parts. A *vision-language model* encodes images and text into a sequence of vectors. An *action expert*, a smaller transformer, turns those vectors into the 50 commands. The action expert works by *flow matching*: it starts from 50 random Gaussian vectors and refines them in 10 steps into the command sequence, each refinement conditioned on the encoded observation. “Flow step $t=1.0$ ” below means the first of those refinement steps.

1.2 Episodes, success, failure

An **episode** starts from one of 50 fixed initial scene configurations (“initial states”, which differ in where the book and the caddy stand), optionally with a *perturbation* applied (the book rotated, the book shifted, or the camera tilted), and runs for up to 520 control steps. The simulator reports **success** if the book ends up in the caddy’s back compartment; otherwise the episode is a **failure**. On this task, most failures are not dramatic: the policy stalls or mis-grasps and the clock runs out. For each failed episode a fixed rule also assigns a failure *onset* step (when the grasp was missed, the book dropped, or the book left the reachable workspace), used only for the lead-time analysis.

1.3 Three kinds of information about a running episode

At each sampled step t (every 5 control steps) we compute a feature vector x_t , and a classifier f_m predicts the probability $p_m(t) = f_m(x_t)$ that *this episode will end in failure*. The three models differ only in what goes into x_t ; the classifier family and its hyper-parameters are identical (gradient-boosted decision trees, fitted once on the Calibration split and frozen).

Model	What it is allowed to see	Concretely (number of features)
M0	Only the policy’s outputs. Nothing about the world beyond what the policy itself computes. This is what a monitor on a real robot could have.	(13) How much the planned commands change when the input is slightly altered—image brightness $\times 0.85$ or $\times 1.15$, the camera yawed $\pm 3^\circ$, the book rotated $\pm 15^\circ$ or shifted a few cm in the simulator and re-rendered; how much the commands vary across 8 random draws of the flow-matching noise; the size and roughness of the planned command sequence.
M1	M0 plus the simulator’s ground truth. Exact poses of gripper, book and goal; contact flags; task phase. Also “coverage”: how close the current state is to states in which Calibration episodes succeeded. A real robot does not have this; it is the strongest <i>non-internal</i> information we can give a monitor.	M0 + relative gripper–book and book–goal poses, symmetry-aware relative yaw, gripper opening, contact flags, phase (pre-grasp / grasped / transport / placed), normalised time; median distance to the 5 nearest successful same-phase Calibration states, distance to the nearest state, failure fraction among the 25 nearest.
M2	M1 plus readouts from inside the network. Everything derived from the activations at one chosen location, through the orientation probe of Section 1.4. Uses no simulator ground truth of its own.	M1 + (8) the probe’s decoded orientation and how it moves under the same input alterations as in M0 (does the readout rotate when the book is rotated? does it wobble under brightness or camera changes or across noise draws?); the readout’s norm and how unusual it is compared with successful Calibration states; and two “controllability” numbers from small test edits inside the probe subspace.

The nesting is the point. M2 vs M1 asks whether the internals contain failure-relevant information that *no* non-internal source has. M2 vs M0 asks whether they contain information a deployed monitor could not otherwise get.

1.4 Reading a variable out of activations: the probe

Pick one location inside the network—here the 720-dimensional vector entering the fourth layer of the action expert at the first flow step, averaged over the 50 command positions—and call it $h \in \mathbb{R}^{720}$. A **linear probe** is a ridge regression from h to a target variable. Our variable is the yaw angle θ_{rel} between the gripper and the book. Because the book looks the same rotated by 180° , the target is the doubled angle on the unit circle,

$$y = (\cos 2\theta_{\text{rel}}, \sin 2\theta_{\text{rel}}) \in \mathbb{R}^2, \quad \hat{y} = Wh + b, \quad W \in \mathbb{R}^{2 \times 720},$$

fitted by $\min_{W,b} \sum_i \|Wh_i + b - y_i\|^2 + \lambda \|W\|_F^2$ on Calibration states and scored by the circular error $|\frac{1}{2} \text{atan2}(\hat{y}) - \theta_{\text{rel}}|$. A good probe means the network *represents* orientation in a way a linear map can extract. Five candidate locations were preregistered; the rule for choosing one and the achieved error are in Section 4.

Key intuition: decodable $\neq$ used

A probe finding θ_{rel} in h shows only that the information is linearly present. The network might read it through the very direction W spans, through some other direction that is merely correlated with it, or not at all (the information could be a by-product). Telling these apart needs an intervention.

1.5 Testing whether the variable is used: patching

The two rows of W span a 2-dimensional subspace of $\mathbb{R}^{720}$; let P be the orthogonal projector onto it (rank 2). Take a *recipient* state with activation h_r and a *donor* state h_d that is the same in every matched respect (task phase, contact, gripper opening, positions) but whose book is rotated by 30–90°. Replace the recipient’s activation by

$$h'_r = h_r + \alpha P(h_d - h_r), \quad \alpha \in \{0.25, 0.5, 1.0\},$$

i.e. move only the probe coordinates of h_r a fraction α of the way toward the donor’s, leave the other 718 directions untouched, and let the network finish computing its command. If the network reads orientation through P , the resulting command should change in the direction appropriate for the *donor’s* orientation (the “target” component), and not much elsewhere (the “off-target” components). Three controls make this a fair test:

- **Random directions.** Repeat the edit with 1 000 random rank-2 subspaces Q in place of P , scaled to the same shift norm. If P is special, its effect must exceed the random distribution (we use its 95th percentile).
- **Matched donor.** Use a donor whose orientation differs by less than 5°. The edit should then do nothing.
- **Staying on the data manifold.** Measure the mean distance from h'_r to its 5 nearest natural Calibration activations; if it exceeds the 95th percentile of that distance among natural states (3.89), the edit has pushed the network into a region it never sees and the result may be meaningless.

1.6 How predictions are scored

Each episode e has a label $y_e \in \{0, 1\}$ (1 = failed) and predictions $p_m(t)$ at the *primary* steps $T_e = \{0, 50, 100, 150, 200\} \cap \{\text{steps before the episode ended}\}$. The episode’s log loss under model m is

$$\ell_e^m = \frac{1}{|T_e|} \sum_{t \in T_e} -[y_e \log p_m(t) + (1 - y_e) \log(1 - p_m(t))],$$

so every episode carries the same total weight regardless of its length. The **primary estimand** is the paired mean difference and its relative version,

$$\Delta = \frac{1}{n} \sum_e (\ell_e^{M2} - \ell_e^{M1}), \quad \text{lift} = \frac{-\Delta}{\frac{1}{n} \sum_e \ell_e^{M1}},$$

with $n = 158$ valid episodes; the preregistered bar was $\text{lift} \geq 3\%$. Two secondary scores: the **Brier score** $\frac{1}{|T_e|} \sum_t (p_m(t) - y_e)^2$ (lower is better), and the **AUROC**, the probability that a randomly chosen failed episode receives a higher predicted failure probability than a randomly chosen successful one (0.5 = chance, 1 = perfect ranking).

Uncertainty. Episodes that start from the same initial state are not independent (same scene, different perturbations), so all intervals are *cluster bootstraps*: resample the 20 initial states of the Locked Test with replacement, keep all 8 episodes of each drawn state, recompute the statistic, repeat 10 000 times, and report the 5th and 95th percentiles (a 90% interval). This is the honest sample size: 20 clusters, not 158 episodes and certainly not 9 988 sampled steps.

Alarms and lead time. A model raises an alarm the first time $p_m(t)$ exceeds a threshold at three consecutive samples. The thresholds were set on Calibration so that at most 10% of *successful* episodes ever alarm. Lead time is the number of control steps between the alarm and the annotated failure onset; we compare the paired lead of M2 and M1 on the failed episodes.

2 Why this question

Interpretability on VLA models has produced decodable representations and steering demonstrations, but rarely a controlled test of whether reading the internals adds *predictive* value over cheaper,

non-internal diagnostics. That is the practical bar: a safety monitor built on activations must beat one built on the policy’s own outputs and, ideally, one built on privileged information nobody has at deployment.

We designed the test as a nested comparison so that a positive result would be unambiguous and a negative result would still be informative. The “boring” alternatives were built in as baselines rather than tested afterwards:

- If internals only beat outputs ($M2 > M0$, $M2 \approx M1$), they *substitute* for state you would otherwise need a simulator for—useful, but not new information.
- If internals beat privileged state ($M2 > M1$), they carry information about the policy’s own competence that the world state does not.
- If a probed direction is causally used, moving it should move actions more than random directions of the same size do, and in the predicted direction.

Before touching the GPU we wrote down (`start.md`, `PREREG.md`) the estimands, the bars, the split logic, and a decision table mapping every outcome to a consequence. Our own toy-model evidence predicted the outcome we then observed: $M2 \approx M1 \gg M0$ (Appendix A).

3 Claims

All four claims below were preregistered as tests; the tiers are ours.

Claim 1 (systematic, preregistered primary): internal signals do not improve failure prediction over privileged state.

Paired log-loss lift $M2$ vs $M1 = +0.47\%$, 90% CI $[-1.2\%, +2.1\%]$ (Δ log loss -0.0024 , CI $[-0.0111, +0.0063]$), bar $\geq 3\%$. **Fails.** Evidence: Section 5.

Claim 2 (systematic, preregistered secondary): internal signals substitute for privileged state.

$M2$ vs $M0$: Δ log loss -0.120 , CI $[-0.187, -0.052]$; relative lift 18.8%. **Holds.** Section 5.

Claim 3 (hedged, preregistered): the probe subspace is not a causal lever at the tested layer and doses.

Sign-correct 48% [36%, 62%]; off-target ratio 4.03 (bar ≤ 0.25); 60% of patched states leave the natural activation manifold. **Specificity fails at** $\alpha \in \{0.25, 0.5, 1.0\}$. The stronger multi-layer criterion could not be evaluated (Section 6).

Claim 4 (narrow, preregistered): no lead-time gain.

Median paired lead-time difference $M2 - M1 = 0$ steps, CI $[0, 0]$, bar ≥ 5 . **Fails.** Section 7.

4 Setup

Policy and task. `lerobot/smolvla_libero` (revision 31d453f7), a SmolVLA policy with a SmolVLM2-500M backbone and a flow-matching action expert (10 denoising steps, 50-step action chunks, one executed action per step). Simulator: LIBERO-10 task 5, “pick up the book and place it in the back compartment of the caddy”; primary object `black_book`, planar symmetry order 2. Two 360×360 cameras (agent view, wrist), 8-D proprioceptive state, 7-D relative actions.

Episodes and failure. Closed loop, replanning every step, horizon 520 control steps after 10 settle steps. Success is LIBERO’s own `check_success`; failure is no success by step 520. A deterministic, frozen annotator assigns each failed episode a failure *onset* step (missed grasp, dropped object, irrecoverable workspace exit, or terminal horizon), used only for lead time.

Perturbation conditions. Eight cells of 20 episodes each, one per initial state: unperturbed (iid); the book rotated by $-37.5^\circ, -22.5^\circ, +22.5^\circ, +37.5^\circ$; the book displaced 4.5 cm diagonally; the agent camera yawed $\pm 5^\circ$ with a 2 cm lateral shift. The Calibration split used different rotation angles ($\pm 15/30/45^\circ$) and a 3 cm axis displacement, so Locked Test conditions are held-out perturbations, not just held-out seeds.

Splits. Fifty fixed initial states: 0–9 Discovery (stack reproduction, reality gate), 10–29 Calibration (all fitting and selection), 30–49 Locked Test. Reset seeds are derived from split, init, and cell; nothing crosses splits. Uncertainty is always a cluster bootstrap over the 20 init-state clusters (10 000 replicates, 90% percentile intervals, seed 260803), never over frames.

The three predictors. All three are histogram-gradient-boosting classifiers (learning rate 0.03, 200 trees, 7 leaves, minimum 20 samples per leaf). The family and hyper-parameters were chosen on Calibration by grouped 5-fold out-of-fold M1 log loss over a small preregistered grid (logistic regression with $C \in \{0.01, \dots, 100\}$ or boosting), then frozen; a Platt calibrator was fitted on the out-of-fold predictions. What each model sees is defined in Section 1: M0 13 output-only features, M1 those plus simulator state and coverage, M2 those plus 8 internal features. No M2 feature uses simulator ground truth.

The probe. As defined in Section 1.4. Five preregistered candidate locations (VLM residual at layer 12; action-expert residual after layers 3 and 11, at flow steps $t=1.0$ and 0.5). Selection rule: episode-weighted ridge regression, grouped 5-fold by init, circular MAE, lowest mean then preregistered preference within one SE. Selected: `early_expert_t1_0` (width 720, mean over the 50 action tokens), MAE 0.253 rad vs. 0.332 for a constant predictor and 0.315 for proprioception alone. The probe is real but modest: it tracks about 15% of a 15° object rotation (median error 12.8°) on Calibration.

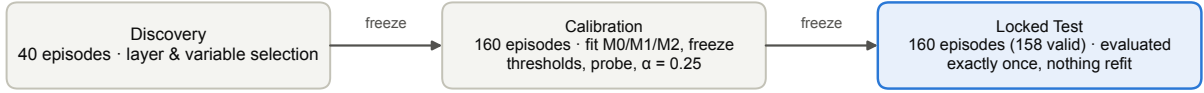
Causal test. The patch and its three controls are defined in Section 1. Donors are matched on phase and contact, gripper opening within 1 cm, gripper and book positions within 2 cm, normalised time within 0.10, and book orientation $30\text{--}90^\circ$ apart; the patch is applied to all 50 command positions at the selected layer and flow step, and the policy’s first command is read. Sixty pairs (3 outcome-blind seeds $\times$ 20), α frozen at 0.25 on Calibration (smallest of $\{0.25, 0.5, 1\}$ meeting the off-manifold cap), $\alpha \in \{0.5, 1.0\}$ as sensitivity.

Blinding and single evaluation. Locked Test labels were never read by a human or by any fitting code before the evaluator ran; the run was monitored by counts and hashes only. Two loader defects (a trailing newline in a frozen lock file; a read-only mapping type) aborted the evaluator at input validation before any metric existed; each was fixed, tested, and recorded as an amendment before the rerun (Appendix D).

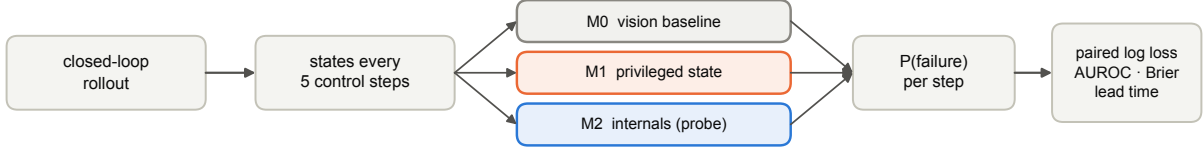
5 Claims 1 and 2: internals substitute for privileged state, and stop there

Why this experiment. The primary comparison isolates the incremental value of internals over *everything* non-internal; the secondary isolates their value over what a deployed monitor could actually see. The four possible outcomes were mapped to consequences in advance (Appendix A).

Splits (LIBERO-10 task 5; 8 condition cells $\times$ 20 init states per split)



Per episode (Locked Test)



labels: terminal failure; primary steps {0, 50, 100, 150, 200}; each episode weight one; 10,000 cluster-bootstrap replicates (20 init clusters)

Causal patching (selected layer `early_expert_t1_0`, 60 pairs $\rightarrow$ 52 valid)

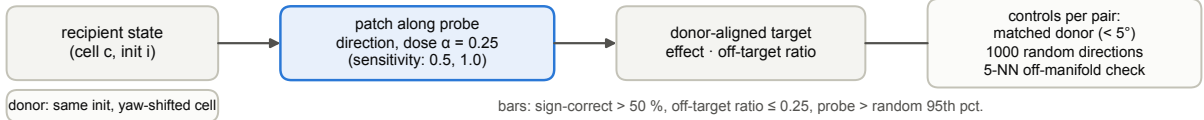


Figure 2: Study design. Three strictly separated splits of 20 initial states each; all fitting, selection and threshold freezing happens on Calibration; the Locked Test is collected only after the freeze and evaluated once. Per episode: closed-loop rollout $\rightarrow$ states every 5 control steps $\rightarrow$ M0/M1/M2 features $\rightarrow$ failure probability per step $\rightarrow$ paired log loss, AUROC, Brier, lead time. Bottom row: the causal test with its three controls.

Result. Table 1 and Figure 1. M1 and M2 are equivalent on every metric; both dominate M0. The primary interval on the relative lift, $[-1.2\%, +2.1\%]$, excludes the 3% bar; its favourable end is 0.9 percentage points short of it. The M2 vs M0 interval excludes zero on the favourable side.

Table 1: Locked Test failure prediction, 158 valid episodes. 90% cluster-bootstrap intervals in brackets. Log loss and Brier: lower is better.

	AUROC	Brier	Log loss
M0 outputs only	0.667 [0.626, 0.710]	0.219 [0.185, 0.256]	0.637 [0.553, 0.727]
M1 + privileged state	0.831 [0.786, 0.876]	0.162 [0.123, 0.205]	0.520 [0.404, 0.649]
M2 + internals	0.835 [0.791, 0.879]	0.162 [0.123, 0.204]	0.517 [0.404, 0.643]
M2 – M1 (paired)			–0.0024 [–0.0111, +0.0063]
M2 – M0 (paired)			–0.120 [–0.187, –0.052]

Per condition. Figure 3. The M1/M2 tie holds in all eight cells. Both are strongest where the perturbation is the probed variable (book yaw $+22.5^\circ / +37.5^\circ$: AUROC 0.93–0.98) and weakest in the iid and camera-yaw cells (0.66–0.73), where failures are rarer and less geometric. M0 is near chance in the iid and camera $+5^\circ$ cells.

Boring explanations we checked.

- *Overfitting on Calibration would hide a real gap.* It does not: the Calibration $\rightarrow$ Locked Test drop is similar for M1 and M2 (AUROC 0.937 $\rightarrow$ 0.831 and 0.938 $\rightarrow$ 0.835; Figure 4), and their Calibration out-of-fold lift was already only 0.40%.
- *Underpowered.* The bar was set at a 3% relative lift; the observed 90% interval has half-width 1.7% and its upper end is 2.1%. A true 3% effect would most likely have shown, but a 1–2% effect is compatible with the data; the preregistered bar, not the point estimate, is what fails.

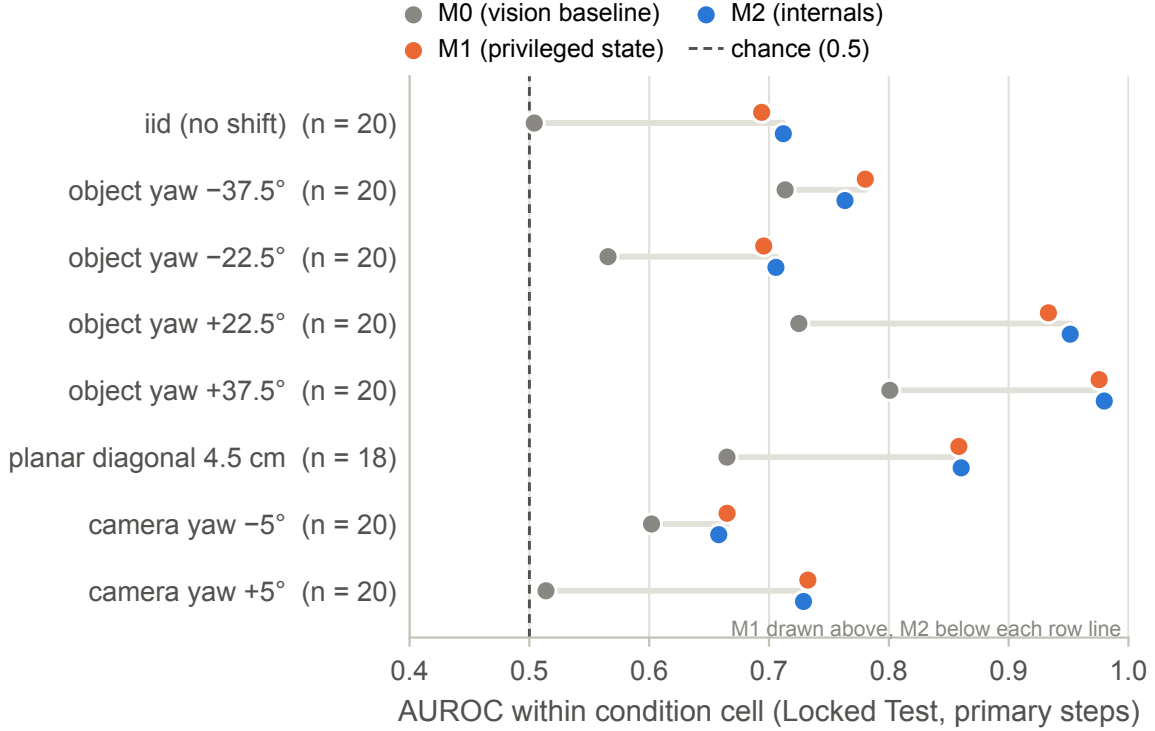


Figure 3: AUROC per perturbation cell (20 episodes each). Dotted line: chance. M1 (orange) and M2 (blue) coincide in every cell; the gap to M0 (grey) is largest under book rotation, the variable the probe encodes.

- *The internal features are broken.* The probe they are built on beats every control on Calibration (circular MAE 0.253 rad vs. 0.332 constant, 0.315 proprioception-only, 0.406 shuffled labels), and removing the M1 coverage features raises the Calibration lift of M2 over M1 from 0.40% to 1.20%—still below the bar (Appendix B). The features carry orientation information; it is simply redundant with the state block.
- *Kill switch.* A preregistered kill switch (any non-internal model with $\text{AUROC} \geq 0.95$ on Calibration would make the task too easy) did not trigger.

Interpretation. On this task the failure-relevant information carried by the orientation probe is a subset of what relative pose and coverage features already encode. That is exactly the “substitute, not exceed” row of the decision table. Note what it does *not* say: it does not say internals are useless at deployment—M2 needs no simulator and beats the output-only monitor by a wide margin.

6 Claim 3: the decodable direction is a readout, not a lever

Why this experiment. A linear probe finding a variable proves decodability, not use. The patching test asks whether *this* 2-D direction is one the policy’s action expert reads from. Possible outcomes: specific and sign-correct (mechanism), sign-correct but unspecific (the direction overlaps something used but is not the used coordinate), or neither (readout only).

Result. Figure 5, Table 2. Of 60 attempted pairs, 52 were valid (eight lacked an eligible donor or matched control; ≥ 30 needed). The patch is small in absolute terms (median donor-aligned effect -1.5×10^{-4} in standardised action units) and, crucially, *smaller than random*: the 95th percentile of the 1000 norm-matched random subspaces is 1.0×10^{-4} , giving a median off-target ratio of 4.03 against a bar of 0.25. The sign of the action change matched the donor’s orientation in 25/52 pairs (48%, CI [36%, 62%]), the matched-donor control—which should be inert—moved the action in the “predicted” direction 54% of the time, and only one of three seeds was positive. Sixty percent of

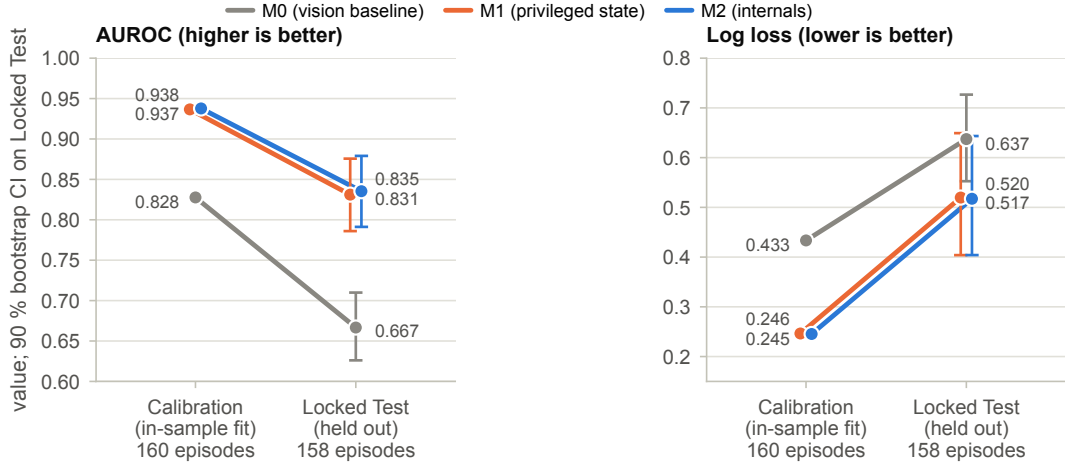


Figure 4: Generalisation gap. AUROC and log loss on Calibration (out-of-fold, where all selection happened) versus the held-out Locked Test with new perturbation angles. All three models lose accuracy; the ordering and the M1/M2 tie are unchanged.

patched states fell outside the natural activation manifold (5-NN distance above the Calibration 95th percentile), so the small effects are not explained by an overly cautious dose.

Table 2: Selected-layer causal patching, $\alpha = 0.25$, 52 valid pairs.

Criterion	Bar	Measured
Sign-correct rate (90% CI)	CI above 50%	48.1% [35.6%, 62.2%]
Median off-target ratio	≤ 0.25	4.03
Target effect vs random p95	target > p95	-1.5×10^{-4} vs 1.0×10^{-4}
Matched-donor ($< 5^\circ$) sign rate	$\approx 50\%$ (inert)	53.8%
Off-manifold rate	informational	59.6%
Seeds with positive direction	$\geq 2/3$	1/3
Supporting layers agree	$\geq 2/3$	unavailable

Dose. Figure 6: at $\alpha = 0.5$ and 1.0, in every one of the eight conditions, the off-target ratio stays between 1.6 and 8.0 and sign-correct rates scatter from 0% to 75% with 4–15 valid pairs per cell—no cell approaches the specificity bar and no dose trend appears (the $\alpha = 0.25$ row, derived from the main evidence, looks the same). Calibration had already shown the same pattern at all three doses (ratios 3.6–5.5), which is why α was frozen at the smallest value rather than tuned upward.

What could not be tested. The preregistered *confirmatory* mechanistic claim required the direction to agree across at least two of three candidate layers. The frozen probe artifact stores coefficients only for the selected layer; fitting the others after Locked Test access would have been a post-hoc refit, so this criterion is reported as *unavailable* and the confirmatory claim as *unsupported*, exactly as decided before access (Appendix D). The single-layer evidence above stands on its own.

Interpretation. The probe direction is decodable because orientation information is present, but the action expert does not read it through this coordinate at this layer—moving it is indistinguishable from noise. Two readings remain open: the used coordinate is elsewhere (different layer, token, or a non-linear function of several directions), or orientation enters the action through the visual pathway in a way that a residual-stream patch at one layer cannot reproduce. Section 8 gives our odds.

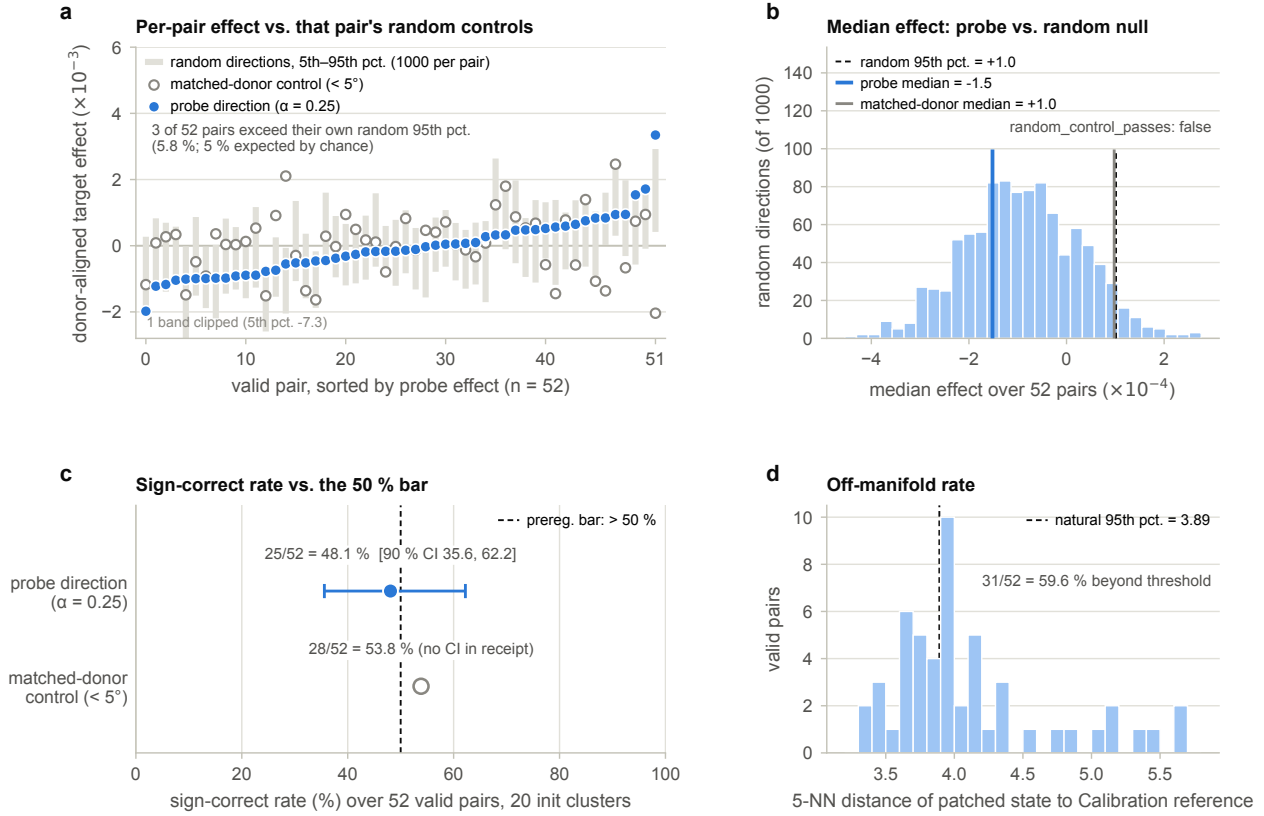


Figure 5: Causal patching at the selected layer (`early_expert_t1_0`, $\alpha = 0.25$, 52 valid pairs). (a) Per pair, the donor-aligned effect of the probe direction (blue) against that pair's 5th–95th percentile band over 1000 random directions of the same shift norm (grey) and the inert matched-donor control (hollow): 3/52 pairs (5.8%) exceed their own random 95th percentile, the 5% expected by chance. (b) The preregistered null: the median-over-pairs effect for each random direction; the probe's median (-1.5×10^{-4}) sits inside the null, below its 95th percentile ($+1.0 \times 10^{-4}$). (c) Sign-correct rate 25/52 = 48.1% with 90% cluster interval versus the 50% bar; matched-donor control 28/52 = 53.8%. (d) 5-NN distance of each patched state to the Calibration reference; 31/52 exceed the natural 95th percentile (3.89).

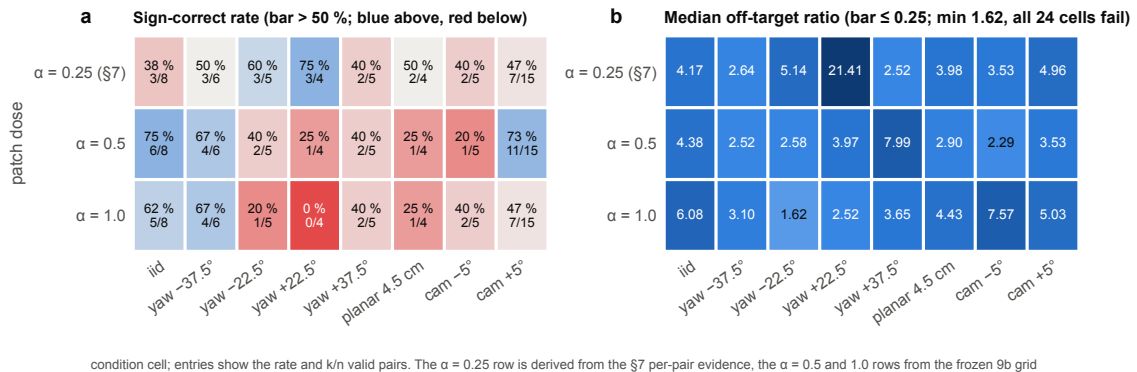


Figure 6: Dose × condition sensitivity. (a) Sign-correct rate with valid-pair counts per cell (grey = 50%, blue above, red below). (b) Median off-target ratio (log scale). Rows $\alpha = 0.5, 1.0$ are the preregistered sensitivity grid; the $\alpha = 0.25$ row is derived from the same pairs in the main causal evidence. Every one of the 24 cells fails the ≤ 0.25 specificity bar (range 1.6–21.4); sign rates scatter from 0% to 75% on 4–15 pairs.

7 Claim 4: no earlier alarm

Thresholds were fixed on Calibration at a 10% episode-level false-positive rate (M0 0.541, M1 0.493, M2 0.504; alarm = 3 consecutive exceedances at 5-step cadence). On the 60 failed Locked Test episodes, M1 and M2 each detect 59; median lead before the annotated onset is 435 and 422.5 steps respectively, and the *paired* median difference is 0 with a degenerate interval $[0, 0]$: in most failed episodes both alarms fire at the same early step. The large absolute leads reflect that most failures on this task are “terminal horizon” failures (on Calibration, 48 of 53) that the state-based models flag almost from the start, not a late, sudden break. Figure 7 shows the paired first-alarm steps and the mean probability trajectories.

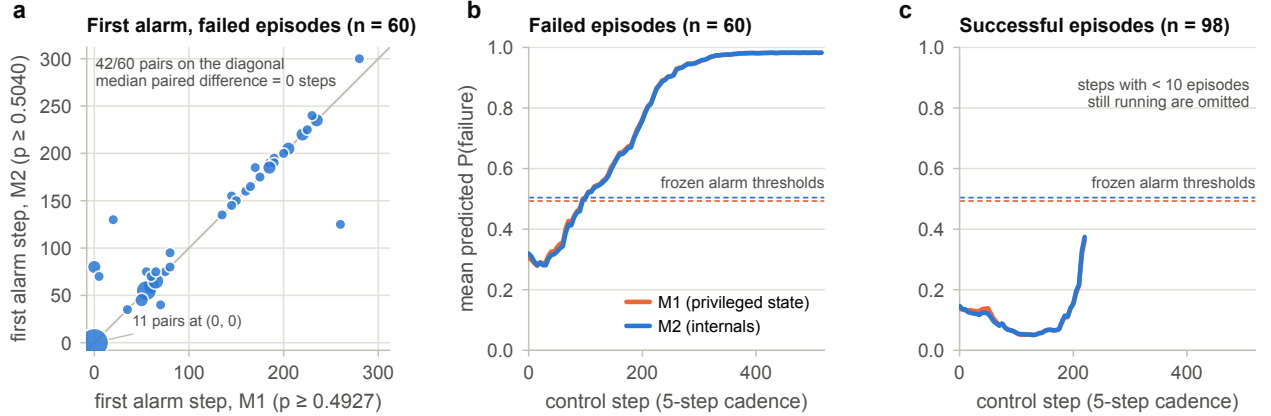


Figure 7: Alarms at 10% episode-FPR. (a) Paired first-alarm control step of M2 versus M1 on the 60 failed episodes (thresholds M1 0.493, M2 0.504); 42/60 pairs lie on the diagonal, marker area $\propto$ coincident pairs; median paired difference 0 steps (bar: ≥ 5). (b, c) Mean predicted failure probability over control steps for failed ($n=60$) and successful ($n=98$) episodes; failed episodes cross the thresholds around step 100 on average, and the M1/M2 curves coincide.

8 Limitations and what would change our mind

We separate three readings of the negative results and give our rough odds; these are judgments, not measurements.

- **Hypothesis false for this policy/task (M2 adds nothing over M1):** $\sim 75\%$. The interval is tight, the Calibration out-of-fold lift was equally small, and the per-cell pattern is uniform. We would update if a different internal feature set (e.g. a probe for gripper–object contact or for the goal compartment) produced a lift $\geq 3\%$ under the same protocol.
- **Method insensitive for the causal claim (wrong layer/coordinate, not absence of mechanism):** $\sim 40\%$. The failure is specific: we tested one linear 2-D subspace at one residual location and flow step, with a patch that is smaller than random noise. We would update toward “mechanism exists” if patching the VLM-context token or the late expert location, or a non-linear (e.g. token-wise) version of the direction, produced a specific, sign-correct effect with the same controls. We would update toward “no mechanism” if a probe with much lower MAE at any location still failed specificity.
- **Underpowered:** $\sim 15\%$ for the prediction claims; $\sim 25\%$ for the causal sign test. The primary interval excludes the 3% bar, but a 1–2% lift would need roughly four times the episodes to resolve. The causal sign interval (± 13 pp) could hide a weak true effect, but the specificity failure does not depend on power.

Further limits: one policy, one task, 158 episodes; failure labels come from the simulator, not human judgment; the annotated failure *onset* is a deterministic heuristic; the probe is modest (tracks $\sim 15\%$ of a rotation); and the study cannot say anything about SAE-style features or non-linear readouts, which were out of scope by design.

What we would do next, if anything. Not another run of this design. The informative follow-ups are (i) a cross-layer causal sweep with the same controls, cheap now that the pipeline exists (≈ 3 GPU-hours per layer), and (ii) replacing the orientation probe with a probe for the variable the policy actually fails on (contact/grasp), which the per-cell results suggest.

9 Reproducibility and cost

Everything is in the repository: preregistration (`PREREG.md`), every change with date and reason (`AMENDMENTS.md`), a per-experiment log (`log.md`), the frozen locks, the runbook, and the final report with all receipts (`artifacts/locked-test-final-report/`). Every input to the evaluator is content-addressed (Table 3); the evaluator refuses to run on anything else.

Compute for the Locked Test: collection 4.3 GPU-h (160 rollouts, ≈ 96 s each), scoring 14.5 GPU-h (the dominant cost: dozens of forward passes per scored state for the counterfactual transformations and noise draws), causal patching 2.8 GPU-h (60 120 forward passes), sensitivity 0.1 GPU-h; 21.7 GPU-h and \$14.81 at \$0.68/h on a rented RTX 5090, plus about \$2 of provisioning overhead. The whole month-one study, including Discovery and Calibration, fits in the low tens of GPU-hours.

Table 3: Content addresses (SHA-256 prefixes) of the frozen inputs and outputs.

Locked Test manifest	1fd8c818
Calibration freeze (probe, predictors, thresholds, α)	eb39e695
Reality-gate lock	4e0d4d5c
Collection receipt (160 artifacts)	3314fb71
Frozen predictions	86d85ed1
Causal receipt (60 evidence files)	5fc67e65
Sensitivity receipt	e6dee93e
Cost receipt	b2547d51
Final report	f7d51bf8

A Preregistered decision table and expectation

From `start.md` §12, fixed on 2 August 2026:

Outcome	Consequence
Lift ($M2 > M1$) and specific patching	Month 2: data-acquisition race
No average lift, clear advantage on delayed failures	Narrow finding; publish
Lift without causal specificity	Good detector, weak mech-interp claim
Specificity without lift	Interesting, economically weak; publish
Neither lift nor specificity	End project, publish negative result
Lift only $M2 > M0$, not $M2 > M1$	Internals replace privileged state, do not exceed it

Pre-stated expectation (`AMENDMENTS.md`, 25 August, from nine toy experiments): “M2 approximately M1 with M2 much better than M0.” Measured: rows five and six.

B Calibration-stage results

Out-of-fold on Calibration (160 episodes, 107 successes): M0 log loss 0.433 / Brier 0.137 / AUROC 0.828; M1 0.246 / 0.068 / 0.937; M2 0.245 / 0.068 / 0.938; M2 vs M1 lift 0.40% (1.20% with the coverage features removed), M2 vs M0 43.4%. Alpha calibration on 60 Calibration pairs: sign rates 56.7% / 60.0% / 51.7% at $\alpha = 0.25/0.5/1.0$ (all intervals contain 50%), off-target ratios 4.94 / 3.63 / 5.51, off-manifold 0%. Probe candidates (circular MAE, rad): VLM context 0.274, early expert $t=1.0$ 0.253 (selected), early $t=0.5$ 0.248, late $t=1.0$ 0.248, late $t=0.5$ 0.249; controls: constant 0.332,

proprioception-only 0.315, shuffled labels 0.406. Discovery reality gate: 8/10 iid successes, 30/30 valid yaw perturbations, failure rate 0.37 (bar 0.20–0.80).

C Locked Test data integrity and per-cell numbers

160/160 artifacts with matching provenance (code commit 18d64941, policy 31d453f7). Two invalid resets (init 47 and 48, cell 5, the 4.5 cm displacement), a 10% rate exactly at the preregistered cap; the cell is retained and the two episodes excluded from all estimands. All other validity-envelope rates are zero.

Cell	Condition	M0	M1	M2
		AUROC / log loss	AUROC / log loss	AUROC / log loss
0	iid	0.504 / 0.716	0.694 / 0.667	0.712 / 0.663
1	book yaw -37.5°	0.713 / 0.537	0.780 / 0.416	0.763 / 0.438
2	book yaw -22.5°	0.566 / 0.492	0.696 / 0.439	0.706 / 0.437
3	book yaw $+22.5^\circ$	0.725 / 0.635	0.933 / 0.487	0.951 / 0.456
4	book yaw $+37.5^\circ$	0.801 / 0.590	0.976 / 0.210	0.980 / 0.204
5	planar diagonal 4.5 cm	0.665 / 0.821	0.858 / 0.700	0.860 / 0.708
6	camera yaw -5°	0.602 / 0.404	0.665 / 0.323	0.658 / 0.316
7	camera yaw $+5^\circ$	0.514 / 0.921	0.732 / 0.934	0.729 / 0.935

D Deviations and amendments during the Locked Test

All recorded prospectively in AMENDMENTS.md before the affected step ran; none touched a scientific quantity.

- **Replacement instance.** The original GPU instance had been destroyed by the provider (negative balance) before the Locked Test; the frozen runtime was rebuilt from pinned wheels, source hashes and model snapshots on a new RTX 5090 (ops/locked_test_bootstrap.sh); the machine-readable preflight matched the freeze on every field.
- **Second scoring worker.** Serial scoring would have exceeded the credit; a second, reverse-walking process was added using the scorer’s built-in concurrency (Calibration precedent showed only physical-cost fields differ).
- **Two evaluator loader fixes** (trailing newline in the frozen reality-gate lock; read-only mapping type from the artifact loader): each aborted the evaluator at input validation with no output, was unit-tested and committed before the rerun. The successful run was the first to read any outcome.
- **Pre-access unavailability decisions** (1 September): the 9a rollout position diagnostic and the supporting-layer coefficients were declared unavailable rather than substituted post hoc; the confirmatory multi-layer claim was therefore fixed as unsupported before the test was opened.

E Glossary

Policy

The neural network that maps an observation (images, joint readings, instruction) to motor commands. Here: SmolVLA, a vision-language model plus a flow-matching action expert.

Episode

One attempt at the task from a fixed initial state, up to 520 control steps; ends in success or failure.

Activation

Any intermediate vector the network computes; we use one 720-dimensional vector at a fixed location.

Probe

A linear regression from an activation to a variable of interest (here the gripper-book yaw), used to test whether the variable is linearly represented.

Patching

Editing an activation during the forward pass and observing the change in output; used to test whether a represented variable is actually used.

Privileged state

The simulator’s exact knowledge of all poses and contacts, unavailable to a real robot.

Counterfactual transformation

Re-rendering the same scene with a small change (brightness, camera yaw, book rotation or shift) and re-running the policy, to measure how sensitive its output is.

Flow matching / flow step

The action expert refines random noise into commands in 10 steps; flow step $t=1.0$ is the first.

Calibration / Locked Test

The split on which everything is fitted and frozen, and the split that is collected only afterwards and evaluated exactly once.

Cluster bootstrap

Resampling whole initial states (with all their episodes) to obtain confidence intervals that respect the dependence between episodes.

Off-target ratio

Size of the output change outside the predicted direction divided by the size along it; a specific intervention has a small ratio.

Off-manifold

An edited activation that is farther from all natural activations than 95% of natural activations are from each other.

F How to reproduce

Clone the repository at tag `locked-test-score-v1`; follow `ops/locked_test_runbook.md`. The report JSON and every receipt are in `artifacts/locked-test-final-report/`; the figures in this document are generated by `make_figures.py` in that directory from those files alone. The raw rollouts (12 GB) and score sidecars are mirrored off-instance and available on request.